Banff National Park Activity Trends

Activity Trends 2020-2025

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1 Land Acknowledgement

Biodiversity Pathways respectfully acknowledges that our work takes place on the territories of Treaties 6, 7, 8, and the Métis homeland, traditional and ancestral lands of First Nations and Métis Peoples, whose histories, languages, and cultures are directly linked to the biodiversity that we monitor.

We acknowledge the traditional teachings of the lands that we work on, and that reciprocal, meaningful, and respectful relationships with Indigenous peoples make our work possible. We are deeply grateful for their stewardship of these lands, and we are committed to supporting Indigenous-led monitoring programs, while learning Indigenous ways of knowing, being, and doing.

2 Introduction

2.1 Overview of NABat and the NNW Bat Hub

North American Bat Monitoring Program (NABat) is a large-scale coordinated effort to monitor bat species to address gaps in knowledge and lack of long-term studies across North America (Loeb et al. 2015).The program is administered by the US Geological Survey (USGS) and implemented by the North by Northwest Bat (NNW) Hub in Alberta, British Columbia, and S.E Alaska.

NNW Bat Hub was established in 2024 with collaboration from Government of Alberta, Government of British Columbia, and Government of Alaska. The hub was formed by combining the previous Alberta Hub and the BC and southeast Alaska Hub. The implementation of the program within Alberta has key support from Parks Canada, government of Alberta staff, other wildlife biologists, Indigenous Nations, community members and naturalists across the province.

2.1.1 NABat Monitoring in Banff National Park

Banff National Park has been conducting NABat monitoring within one NABat grid cell (Grid Cell ID: 148842) since 2020. Monitoring efforts include three long-term stationary sites and a driving transect. In addition, 10 supplementary sites were surveyed in 2022 and 2023 to develop a bat species inventory for the park. All data from 2020 to 2023 were processed using automated identification software. In 2024 and 2025 a subset of the data were also manually verified following the established NABat protocols (Reichert et al. 2018a).

2.2 Threats to bat populations in Alberta

Bat populations in Alberta face a wide range of threats including forestry practices, anthropogenic expansion, and climate change among others. Of these threats, White-Nose Syndrome and wind energy development have emerged as two of the highest-priority concerns.

2.2.1 White-Nose Syndrome

White-Nose Syndrome (WNS) is a deadly fungal disease caused by the fungus *Pseudogymnoascus destructans* (*Pd*) that affects hibernating bats. As of 2026, WNS has been confirmed in nine Canadian provinces including Alberta (Figure 1: CWHC-RCSF (2024)). Since Pd was first confirmed in AB in 2022, the fungus and the ensuing disease have spread through the province at a rapid pase and has now been detected at the provinces largest known hibernacula, Cadomin cave (Wilkinson, pers. comm.) Continued monitoring of populations in AB is critical as *Pd* and WNS continue to spread throughout the province.

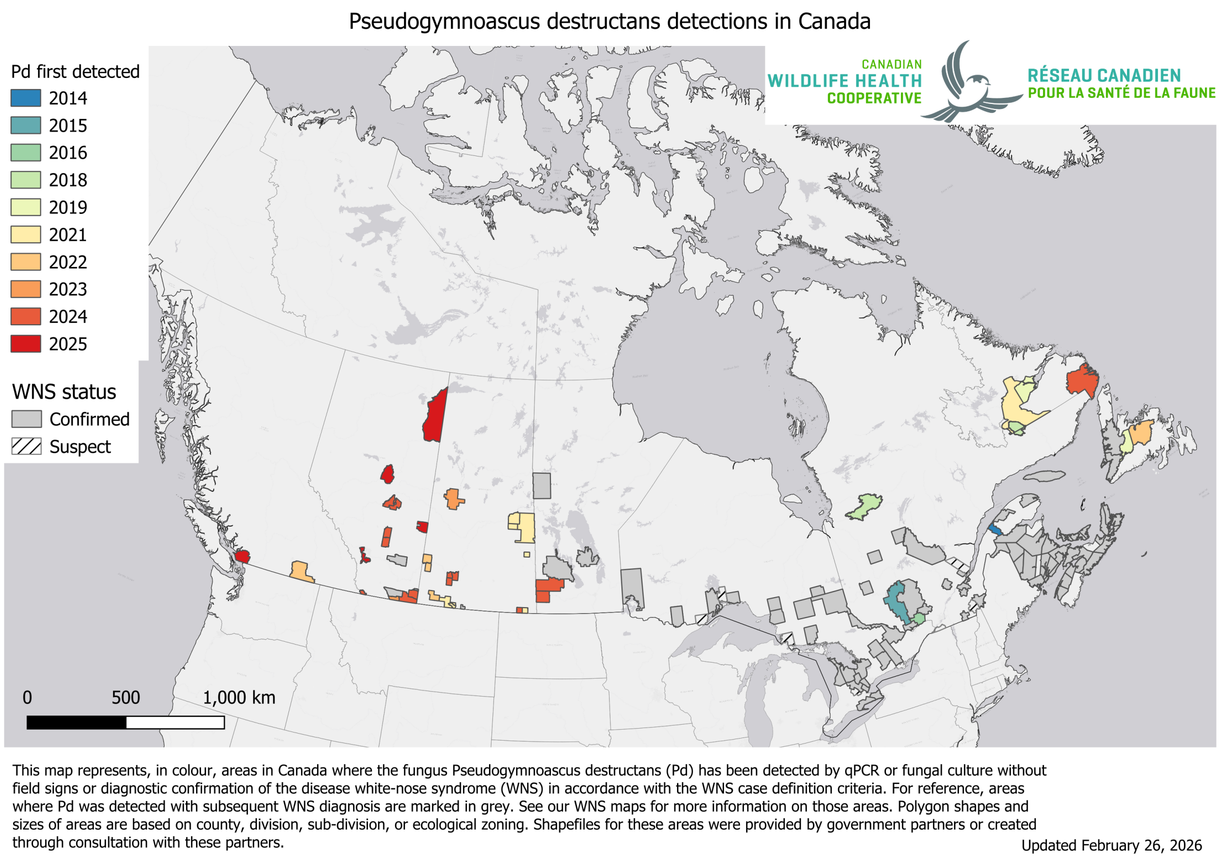


Figure 1: White-nose syndrome (WNS) and Pseudogymnoascus destructans (Pd) occurrence in Canada. Source: CWHC-RCSF (2026)

2.2.2 Wind Energy Development

Wind energy farms have been linked to high mortality rates across all bat species in Canada, with long distance migratory species accounting for 73% of recorded fatalities (Zimmerling and Francis 2016). In Alberta alone, one estimate predicts an annual mortality rate 10.9 bats per turbine (Zimmerling and Francis 2016). There are currently 1,778 turbines active in the province across 49 different projects with the most turbines found in the grassland ecosystem ("Canadian Wind Turbine Database" (2025) ; Figure 2). Using Zimmerling and Francis (2016) estimate, Alberta is estimated to be loosing 19,380 bats per year to wind turbine collisions. As of 2023, COSEWIC designated all three of Alberta's long distance migratory species as endangered due in part to fatalities at wind turbine facilities (Status of Endangered Wildlife in Canada 2023).

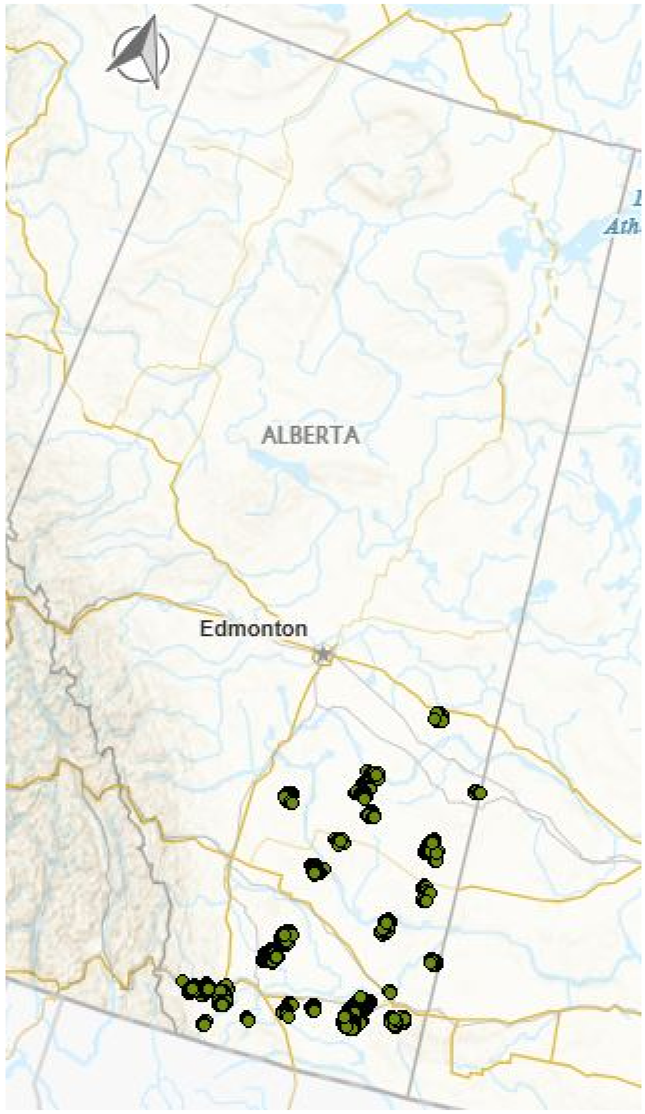


Figure 2: Current wind turbines in Alberta based on the Canadian Wind Turbine Database (Source: "Canadian Wind Turbine Database" (2025))

2.3 Acoustic Monitoring of Bats

Acoustic recording is a cost-effective method for monitoring bat populations across large geographic and temporal scales. However, several limitations affect the conclusions that can be drawn from these data:

1. **Imperfect species identification** — Bats rely on echolocation for hunting and moving around, and species with similar ecology and physiology often produce overlapping calls. While detectors are deployed to maximize recordings of diagnostic calls, species identifiability will always vary depending on the local species assemblage.

2. **Variable detectability** — Bat species are not equally detectable acoustically. Larger, wide-ranging species produce louder calls that travel farther, while smaller species with quieter calls or specialized habitat use are less reliably recorded. As a result, trend estimates cannot be produced for all species using acoustic methods alone.
3. **Data variability** — Many factors influence recording quality and bat activity, including wind speed, humidity, and local habitat. Additionally, stationary detectors cannot account for individual recapture, meaning acoustic data can only inform relative activity and occupancy.
4. **Processing limitations** — Auto-ID software enables efficient processing of large datasets but has known high misclassification rates for bats, making manual verification necessary to ensure accurate results.

3 Methods

3.1 *Deployments*

3.1.1 Stationary Surveys

Banff biologists deployed three stationary bat detectors in strategically selected bat habitat across distinct quadrants within the NABat grid cell (GRID Cell ID: 148842; Figure 3). These stationary sites were deployed consistently every year from 2020 to 2025, with detectors set out between June 24–30 and retrieved between July 2–23 (Table 1). The comparable deployment windows across years allow for interannual comparison. Deployment setup followed the standards established by Loeb et al. (2015). Titley Swift units were used in most years, except in 2020 and 2024, when Wildlife Acoustics SM4 units were deployed instead.

3.1.2 Transect Surveys

The driving transect protocol closely follows that established by Loeb et al. (2015). Between one and three replicates of a mobile transect were driven at 30–35 km/h with a bat detector mounted on the vehicle roof to record bats along the road (Figure 3). Transects were conducted during the same period that the stationary acoustic units were deployed (Table 1). Titley Walkabout units were used for all transect deployments across all years.

Table 1: Deployments throughout the 6 years of this monitoring

Year	Detector	Start	End	Nights
Fenlands				
2020	Wildlife Acoustics SM4	27 Jun	08 Jul	11
2021	Titley Swift	29 Jun	15 Jul	16
2022	Titley Swift	24 Jun	09 Jul	15
2024	Wildlife Acoustics SM4	26 Jun	08 Jul	12
2025	Titley Swift	24 Jun	02 Jul	8
Golf Course				
2021	Titley Swift	29 Jun	15 Jul	16
2022	Titley Swift	24 Jun	09 Jul	15
2023	Titley Swift	28 Jun	12 Jul	14
2024	Wildlife Acoustics SM4	26 Jun	08 Jul	12
2025	Titley Swift	24 Jun	02 Jul	8
Upper Hotsprings				
2021	Titley Swift	29 Jun	15 Jul	16
2022	Titley Swift	26 Jun	09 Jul	13
2023	Titley Swift	28 Jun	12 Jul	14
2024	Wildlife Acoustics SM4	26 Jun	08 Jul	12
2025	Titley Swift	24 Jun	02 Jul	8
Driving Transect				
2021	Titley Walkabout	06 Jul	06 Jul	1
2022	Titley Walkabout	24 Jun	24 Jun	1
2022	Titley Walkabout	27 Jun	27 Jun	1
2022	Titley Walkabout	30 Jun	01 Jul	1
2023	Titley Walkabout	28 Jun	29 Jun	1
2023	Titley Walkabout	30 Jun	30 Jun	1
2024	Titley Walkabout	30 Jun	30 Jun	1
2024	Titley Walkabout	01 Jul	01 Jul	1
2024	Titley Walkabout	02 Jul	02 Jul	1
2025	Titley Walkabout	24 Jun	24 Jun	1
2025	Titley Walkabout	01 Jul	01 Jul	1

3.2 *Manual Verification*

All recordings were classified using the Kaleidoscope Pro Bats of North America 5.4.0 classifier or newer. Automatic classification was used to assign species identifications and to exclude non-bat recordings. Manual verification of classifications was conducted for the 2024 and 2025 seasons only, following the processing standards established by the North American Bat Monitoring Program (NABat) (Reichert et al. 2018b). Only recordings assigned a species classification by Kaleidoscope were selected for manual verification. For stationary acoustic monitoring, recordings were manually vetted until at least one recording per species per site per night was confidently identified. For mobile transects, all recordings with automated species classifications were manually vetted. Species identifications were validated against reference call parameters described by Slough et al. (2022), Solick (2022), and Szewczak (2018), adhering to NABat manual vetting standards (Reichert et al. 2018b).

Based on documented species ranges and prior detection data (Olson, n.d.), manual identification efforts focused on seven species: Big Brown Bats (*Eptesicus fuscus*), Eastern Red Bats (*Lasiurus borealis*), Silver-haired Bats (*Lasionycteris noctivagans*), Hoary Bat (*Lasiurus cinereus*), Little Brown Bats (*Myotis lucifugus*), Long-legged Myotis (*Myotis volans*) and Long-eared Myotis (*Myotis evotis*).

Trends estimates used only autoID results unless the manual verification results indicated a bat recordings was actually noise. This was done to maintain consistency across the 6 years of study, since manual verification only started in 2024.

All the recordings and results can be found on Widltrax under the Banff National Park Organization, in the project called Bat Community - Long Term Monitoring - NABat - 2020-2025.

3.3 *Statistical Analysis*

3.3.1 *Species summaries*

Species presence and nightly activity were summarized from a site–night–species table constructed for all sampled nights. Sampling effort was defined as the number of unique site-nights on which a detector was recording. Mean nightly passes include all recordings assigned a species or couplet label; couplet-labelled files were counted toward both member species.

3.3.2 *Environmental covariates*

Weather data were obtained from the nearest Environment and Climate Change Canada (ECCC) stations using the `weathercan` package (Version 1.0.0) in R and summarized for each sampling night as mean temperature, mean relative humidity, total precipitation, and mean wind speed. Moon covariates were calculated for each site-night using the `suncalc` package (Version 0.5.1) and included moon illumination and an index of moon position. All environmental covariates were standardized (z-scored) prior to modelling to aid convergence and comparability. The year term was left on its original scale so that trend estimates represent change per year.

3.3.3 *Trend models*

3.3.3.1 Data included

Trend models used stationary detector data only. Transect data were not modelled because the number of recorded call sequences was too low to support abundance estimation, which requires substantially more sites (grid cells) and transect replicates than were available here. These data will nonetheless be useful for modelling species abundance at provincial or regional scales.

3.3.3.2 Species

Trends were estimated for six taxa: Big Brown Bat (*Eptesicus fuscus*, EPFU), Eastern Red Bat (*Lasiurus borealis*, LABO), Hoary Bat (*Lasiurus cinereus*, LACI), Silver-haired Bat (*Lasionycteris noctivagans*, LANO), Long-eared Myotis (*Myotis evotis*, MYEV), and a 40 kHz Myotis complex (40KMyo) comprising *M. lucifugus*, *M. volans*, and *M. septentrionalis,* whose recordings were summed within each site-night prior to modelling. We used the 40kHz Myotis complex due to the high proportion of error by Kaleidoscope and differentiating between these species. The Long-eared Myotis is not included in the 40KMyotis complex due to it's lower frequency call (Fc~30-35kHz) making it more identifiable, and less likely to be misidentified by the program.

3.3.3.3 Model structure and selection

Nightly bat activity (number of recordings per site-night) was modelled using negative binomial generalized linear mixed models (GLMMs) with a log link, fitted in `glmmTMB` (Version 1.1.12). Each model included a random intercept for deployment (`1 | site:year`). This term was used in preference to a site-only intercept because nights within a deployment share a common detector, configuration, and weather window, and are therefore not independent replicates of a year. Fixed effects comprised a linear year term (years since 2020), a two-level detector-unit term (Wildlife Acoustics SM4 vs. Titley Anabat Swift) included to account for potential differences in detection attributable to unit type, and a taxon-specific set of environmental covariates.

For each taxon, candidate models were ranked by AICc and the covariate set from the top-ranked (lowest AICc) model was retained.

4 Results

Acoustic monitoring happened for a total of 191 detector nights across 3 stationary survey locations and 11 driving transects nights over six years (2020–2025, Table 2). Survey effort was mostly even within sites and years with sites sampled for a mean of 12.7 nights per year (range: 9–16 nights per site-year). A total of 41609 bat detections were recorded across the three stationary sites and a total of 475 where collected during driving transects in the Banff NABat grid cell.

Table 2: Monitoring effort and bat species detections in the Banff (2020–2025). Mean nightly recordings (± SD) represent total bat detections per night summed across species. Species detections indicate presence (X) within each year and quadrant based on combined automated and manually reviewed identifications. 2024 and 2025 are bolded to highlight that these were the only years were manual verification occurred.

Year	Site	Nights	Mean Nightly Bat Recordings ± SD	EPFU	LABO	LACI	LANO	MYEV	MYLU	MYSE	MYVO
2020	Fenlands	11	75.5 ± 118.6	X	X	X	X	X	X		X
2021	Fenlands	16	236.2 ± 330.7	X	X	X	X	X	X		X
2021	Golf Course	14	542.4 ± 150.6	X	X	X	X	X	X		X
2021	Upper Hotsprings	16	7.9 ± 9.9	X	X	X	X	X	X		X
2022	Fenlands	14	335.0 ± 298.5	X	X	X	X	X	X	X	X
2022	Golf Course	13	254.2 ± 128.4	X	X	X	X	X	X	X	X
2022	Upper Hotsprings	14	2.5 ± 2.6			X	X	X	X		
2023	Golf Course	14	452.9 ± 246.5	X	X	X	X	X	X	X	X
2023	Upper Hotsprings	14	5.4 ± 8.4	X	X	X	X	X	X		X
2024	**Fenlands**	**13**	**724.8 ± 468.4**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**
2024	**Golf Course**	**13**	**375.6 ± 178.4**	**X**	**X**	**X**	**X**	**X**	**X**	**X**	**X**
2024	**Upper Hotsprings**	**12**	**4.5 ± 6.0**	**X**	**X**	**X**	**X**	**X**	**X**		
2025	**Fenlands**	**9**	**34.6 ± 14.0**	**X**	**X**	**X**	**X**	**X**	**X**		**X**
2025	**Golf Course**	**9**	**14.2 ± 8.3**	**X**	**X**	**X**	**X**	**X**	**X**		**X**
2025	**Upper Hotsprings**	**9**	**4.1 ± 1.8**	**X**	**X**	**X**	**X**	**X**	**X**		
2021	Driving Transect	1	36.0 ± NA	X	X	X	X		X		
2022	Driving Transect	3	57.7 ± 13.2	X	X	X	X		X		
2023	Driving Transect	2	46.0 ± 7.1	X	X	X	X		X		
2024	**Driving Transect**	**3**	**12.7 ± 0.6**	**X**	**X**		**X**	**X**	**X**		**X**
2025	**Driving Transect**	**2**	**68.0 ± 8.5**	**X**	**X**	**X**	**X**		**X**		**X**

4.1 *Species detected and mean nightly activity*

Eight species were recorded across the study. Silver-haired bat (*Lasionycteris noctivagans*) and Little Brown Bat (*Myotis lucifugus*) were detected across every site/driving transect and year. Northern myotis (*Myotis septentrionalis*) was the least detected species, recorded only at Fenlands in 2022 and 2024, and Golf Course in 2022, 2023 and 2024 (Table 2, Figure 3). The Little Brown Bat had the highest number of species detections across all years (20304 ; nightly mean = 100.5148515). This was followed by Long-legged Myotis (2532 ; nightly mean = 12.5346535), and Eastern Red Bat (1338 ; nightly mean = 6.6237624).

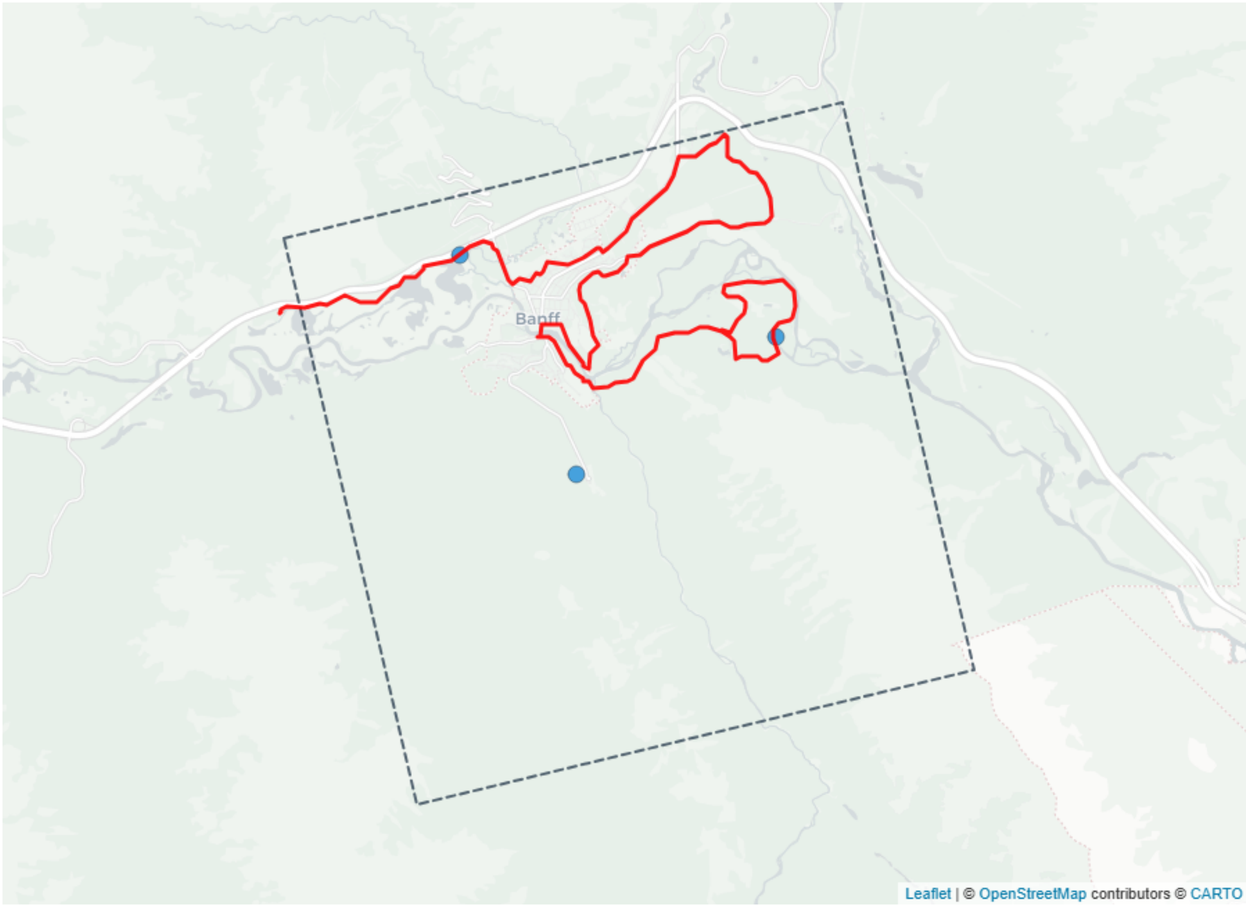


Figure 3: Acoustic bat monitoring sites in Banff. Stationary recording stations are shown as blue circular markers and driving transect is in red line. The boundary of the grid cell is shown in dashed grey line.

4.2 *Species Trends*

Across all six modelled taxa, the year term (YearS, years since 2020) was non-significant, with 95% confidence intervals on every incidence rate ratio (IRR) encompassing 1.0 (all p 0.23). We therefore detected no statistically significant directional change in nightly detection rates over the 2020–2025 monitoring period, consistent with relatively stable bat activity within the Banff grid cell (Table 3).

Table 3: Year-over-year trend estimates for six bat taxa in the Banff NABat grid cell, 2020–2025. Trends are from negative binomial GLMMs of nightly detections and are expressed as incidence rate ratios (IRRs) for the year term. IRR > 1 indicates an increasing trend, IRR < 1 a decreasing trend, and IRR $= 1$ no change. A 95% confidence interval spanning 1.0 indicates the trend is not statistically significant at $= 0.05$. "Covariates Used" lists the environmental terms retained in each taxon's model; a deployment random intercept, year, and detector unit were included in all models.

Species	Covariates Used	IRR (95% CI)	p-value
EPFU	Nightly Mean Temperature	1.39 (0.81–2.36)	0.23
LABO	Nightly Mean Temperature + Nightly Precipitation + Nightly Mean Wind Speed	0.88 (0.49–1.58)	0.67
LACI	Nightly Mean Temperature + Moon Illumination Fraction + Nightly Precipitation + Nightly Mean Wind Speed	0.90 (0.64–1.25)	0.52
LANO	Nightly Mean Relative Humidity + Nightly Mean Wind Speed + Nightly Mean Temperature	0.97 (0.68–1.39)	0.87
40KMyo	Nightly Mean Relative Humidity + Nightly Mean Temperature + Nightly Precipitation	0.70 (0.34–1.46)	0.35
MYEV	Nightly Precipitation + Nightly Mean Temperature + Nightly Mean Wind Speed	1.16 (0.80–1.69)	0.44

Estimates varied in both direction and magnitude. Big Brown Bat (EPFU; IRR = 1.39, 95% CI 0.81–2.36; p = 0.23) and Long-eared Myotis (MYEV; IRR = 1.16, 95% CI 0.80–1.69; p = 0.44) trended positive, while Eastern Red Bat (LABO; IRR = 0.88, 95% CI 0.49–1.58; p = 0.67) and Silver-haired Bat (LANO; IRR = 0.97, 95% CI 0.68–1.39; p = 0.87) were essentially flat to weakly negative (Figure 4). None of these approached significance.

Two taxa showed negative estimates that, while not significant, are important to note. The 40 kHz Myotis complex (40KMyo) had the steepest negative trend, with an IRR of 0.70 (95% CI 0.34–1.46; $p = 0.35$). This estimate corresponds to roughly a 30% decline in nightly activity per year. The Hoary Bat likewise trended negative (LACI; IRR = 0.90, 95% CI 0.64–1.25; p = 0.52), corresponding to an estimate of about a 10% annual decline.

5 Discussion

Our six-year activity-trend analysis for the Banff grid cell revealed no statistically significant directional change in nightly bat activity for any of the six modelled taxa, providing preliminary evidence of relatively stable bat activity within the cell since monitoring began in 2020. Four taxa trended positive or flat: Big Brown Bat (EPFU) and Long-eared Myotis (MYEV) positive, Silver-haired Bat (LANO) and Eastern Red Bat (LABO) essentially flat (Figure 4). Two taxa showed negative point estimates: the 40 kHz Myotis complex (40KMyo) declined by roughly 30% per year and Hoary bat (LACI) by roughly 10% per year. Although neither reached significance, both are taxa of conservation concern, and continued monitoring will be needed to determine whether these apparent declines are real.

These local patterns are best interpreted alongside broader-scale analyses. Province-wide NABat relative abundance modelling has documented a significant ~40% decline in abundance for Hoary

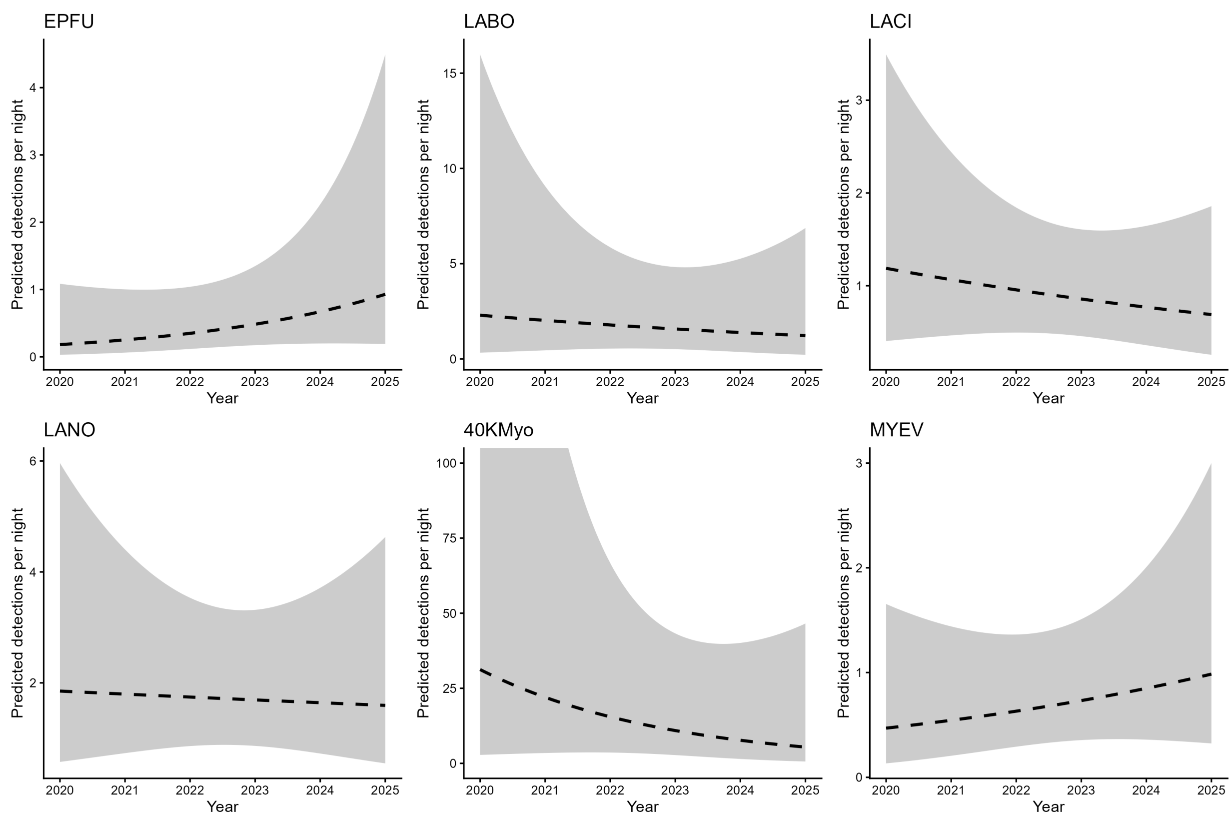


Figure 4: Species relative activity trends from 2020–2025 in the Banff NABat grid cell. Lines show trend estimates of nightly detections with 95% confidence intervals. Dashed lines indicate non-significant trends ($p > 0.05$). Models are based on nightly acoustic detections and fitted using negative binomial mixed-effects models.

Bats across Alberta between 2012 and 2023 (mean = −0.39, 95% CI −0.61 to −0.13), together with a smaller, non-significant decline of ~15% for Silver-haired bats (mean = −0.15, 95% CI −0.45 to 0.22; (Bradley J Udell et al. 2025)). Although our grid-cell estimates were not significant, the negative direction of our Hoary bat trend is consistent with this regional signal. The difference in statistical power is expected: the provincial estimates integrate data across all of Alberta, whereas our analysis is restricted to a single grid cell. Our smaller sample size and spatial extent produce greater variance and reduced power to detect trends, so the absence of local significance does not contradict the broader decline. Placing our results within this regional and continental context is therefore essential to their interpretation. The decline of migratory tree bats such as the Hoary Bat has been widely linked to mortality at wind-energy facilities ((Zimmerling and Francis 2016)). The continued expansion of wind-energy development across the region may therefore place further pressure on this species.

The 40KMyo complex warrants particular attention. NABat occupancy analyses for Alberta (2010–2019) suggest that Little Brown bat, Northern Myotis, and Long-legged Myotis populations were stable over that period ((Udell et al. 2022)). Our complex, which we expect is dominated by the Little Brown bat, instead points to a potential decline, albeit without statistical significance. This divergence is most likely due to the recent arrival of white-nose syndrome (WNS) to the province. *Pseudogymnoascus destructans* (Pd), the causative fungus, was first detected in Alberta in 2022 and has since spread across the province ((CWHC-RCSF 2024)). WNS was confirmed in the south eastern part of the province in 2023, and as of winter 2026 WNS has been confirmed in the Rocky Mountains, including at Cadomin Cave, a major regional hibernaculum (L. Wilkinson, pers. comm.). Given the severe overwinter mortality WNS causes in hibernating *Myotis*, marked declines inLittle Brown bat and Northern Myotis, are anticipated in the coming years. While our study period largely predates confirmed WNS at this location, the negative 40KMyo estimate underscores the importance of sustained monitoring of this complex as the disease establishes.

These conclusions must be tempered by several limitations. Most importantly, our inferences are constrained by the small sample size and limited spatial extent of a single grid cell, which reduce statistical power and amplify the influence of individual site-years. In addition, the trend analysis relied solely on automated classifications from Kaleidoscope, which misclassifies bat calls at high rate. Such misclassification introduces uncertainty into species-level detection counts and may bias trend estimates, particularly among acoustically similar species.

Overall, and with these caveats in mind, our results indicate that bat activity within the Banff grid cell has remained relatively stable over the 2020–2025 monitoring period. However, the negative point estimates for Hoary bat and 40KMyotis comples, combined with documented regional declines and the ongoing spread of WNS and wind turbine development, reinforce the value of continued long-term acoustic monitoring at this site.

6 Acknowledgements

We thank the Parks Canada biologists in Banff National Park who collected the acoustic data. The NNW Bat Hub, which coordinates and compiles long-term acoustic bat monitoring data for the province of Alberta, has been funded in part by the Government of Alberta.

7 Appendix A

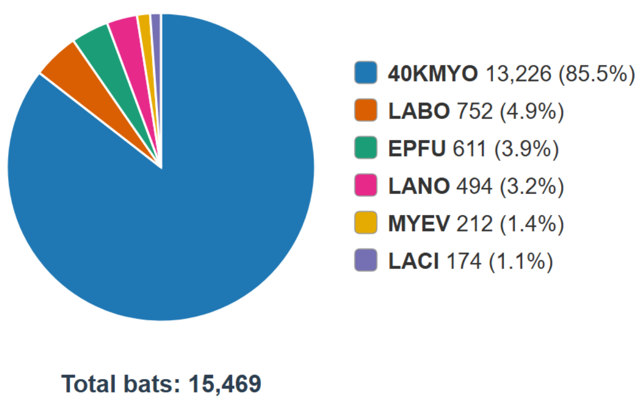
Species codes and their definitions

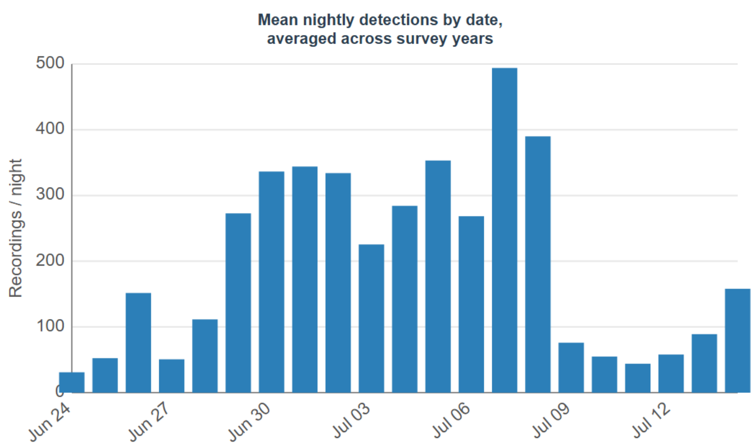
CommonName	ScientificName	Code	Definition
Big Brown Bat	Eptesicus fuscus	EPFU	Calls that have diagnostic features identifying it as Eptesicus fuscus
Big Brown Bat / Silver-haired Bat	Eptesicus fuscus / Lasionycteris noctivagans	EPFULANO	Calls that could be attributed to either Eptesicus fuscus or Lasionycteris noctivagans
Eastern Red Bat	Lasiurus borealis	LABO	Calls that have diagnostic features identifying it as Lasiurus borealis
Eastern Red Bat / Little Brown Myotis	Lasiurus borealis / Myotis lucifugus	LABOMYLU	Calls that could be attributed to either Lasiurus borealis or Myotis lucifugus
Hoary Bat	Lasiurus cinereus	LACI	Calls that have diagnostic features identifying it as Lasiurus cinereus
Hoary Bat / Silver-haired Bat	Lasiurus cinereus / Lasionycteris noctivagans	LACILANO	Calls that could be attributed to either Lasiurus cinereus or Lasionycteris noctivagans
Silver-haired Bat	Lasionycteris noctivagans	LANO	Calls that have diagnostic features identifying it as Lasionycteris noctivagans
Little Brown Myotis	Myotis lucifugus	MYLU	Calls that have diagnostic features identifying it as Myotis lucifugus
Little Brown Myotis / Northern Myotis	Myotis lucifugus / Myotis septentrionalis	MYLUMYSE	Calls that could be attributed to either Myotis lucifugus or Myotis septentrionalis
Northern Myotis	Myotis septentrionalis	MYSE	Calls that have diagnostic features identifying it as Myotis septentrionalis
40kHz Frequency Myotis		40kMyo	Various species of Myotis that have a characteristic frequency in the range of 35-40kHz
High Frequency Bat		HIGH FREQUENCY	Various species with pulses having a characteristic frequency higher than ~35kHz
Low Frequency Bat		LOW FREQUENCY	Various species with pulses having a characteristic frequency lower than ~30kHz
Unknown Bat		NoID	Bat calls but no grouping category applies
Noise		Noise	No bat recorded

8 Appendix B

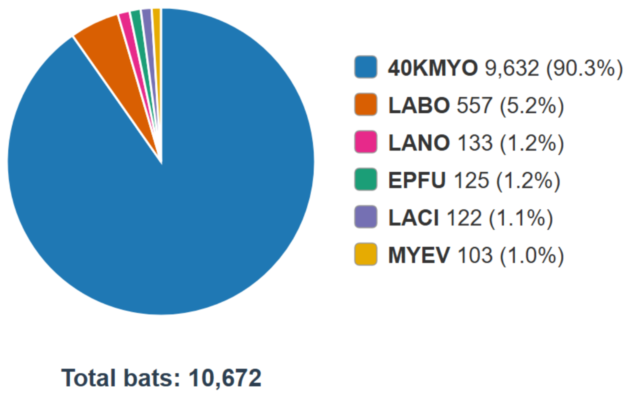
Species detection summaries for each monitoring site.

Fenlands

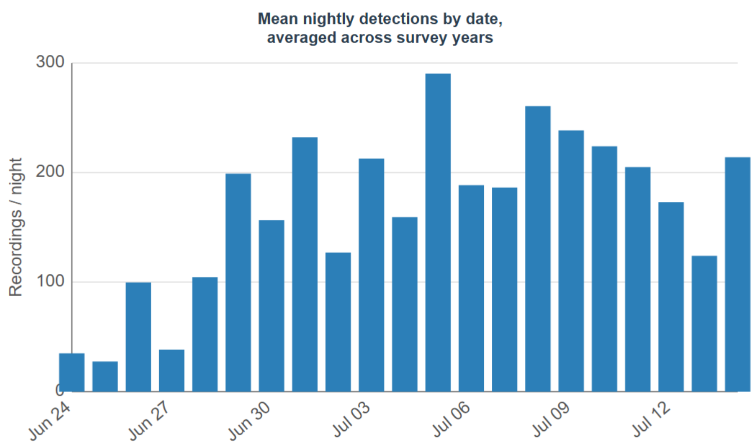




Golf Course

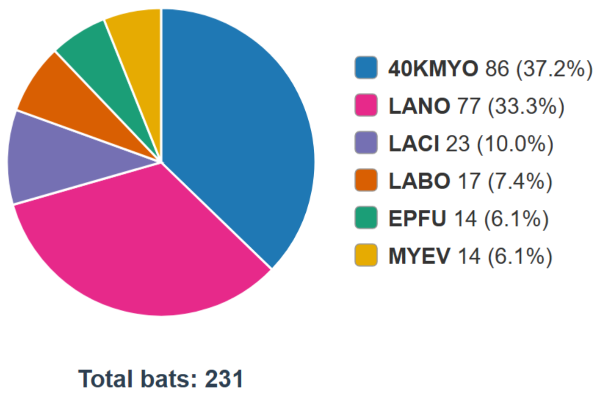


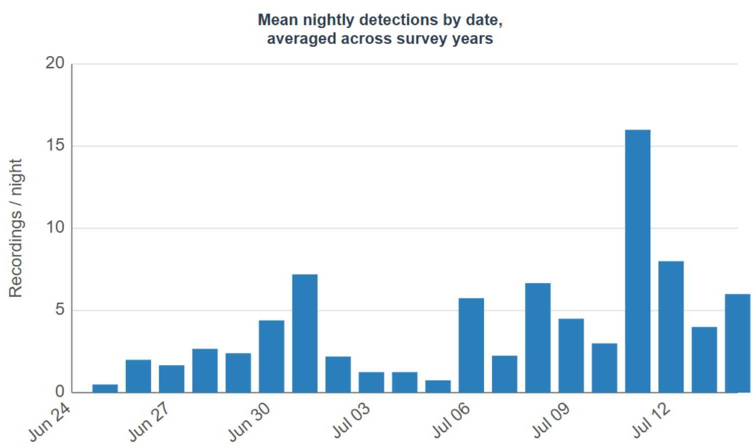
40KMYO 9,632 (90.3%)
LABO 557 (5.2%)
LANO 133 (1.2%)
EPFU 125 (1.2%)
LACI 122 (1.1%)
MYEV 103 (1.0%)
Total bats: 10,672



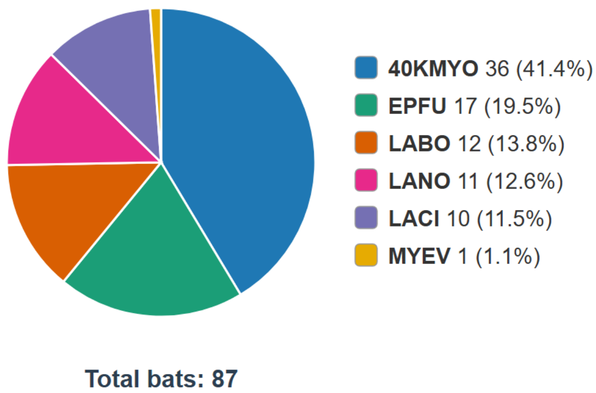
Mean nightly detections by date,
averaged across survey years
Recordings / night
300
200
100
0
Jun 24
Jun 27
Jun 30
Jul 03
Jul 06
Jul 09
Jul 12

Upper Hotsprings





Driving Transect



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